Name	( )	Class	

RIVER VALLEY HIGH SCHOOL
2018 Year 6 Preliminary Examination
Higher 2

MATHEMATICS 9758/02

Paper 2

17 September 2018

3 hours

Additional Materials: Answer Paper
List of Formulae (MF26)
Cover Page

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, class and index number in the space at the top of this page.

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphic calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **7** printed pages and **1** blank page.

Section A: Pure Mathematics [40 marks]

1. A curve C has equation $y=\dfrac{x^2-3x+3}{x-2}$ where $1\le x\le 3$ and $x\ne 2$.

(i) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ and comment on the gradient of C. [3]

(ii) Sketch the graph of C. [2]

(iii) By adding a suitable graph to the sketch of C, solve the inequality

$$\frac{x^2-4x+5}{x-2}\ge 3\left(x-2\right)^2,\ \text{ for } 1\le x\le 3,\ x\ne 2.$$ [3]

(iv) The function g is defined by $\mathrm{g}(x)=\dfrac{x^2-3x+3}{x-2}$ for $1\le x\le 3$ and $x\ne 2$ such that $\mathrm{g}(x)=\mathrm{g}(x+3)$. Sketch the graph of $y=\mathrm{g}(x)$ for $-1<x<5$. [2]

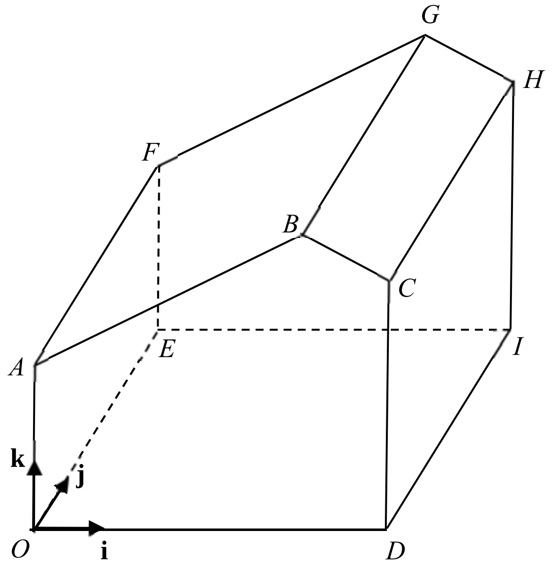
2. **(i)** Express $\dfrac{2(2r+3)}{(r+1)(r+2)(r+3)}$ as partial fractions. [2]

(ii) Hence find an expression for $\displaystyle\sum_{r=1}^{n}\frac{2r+3}{(r+1)(r+2)(r+3)}$ in terms of n. [4]

(iii) Using the result in part **(ii)**, show that the following series is convergent and find its sum to infinity.

$$\frac{21}{3\times4\times5}+\frac{27}{4\times5\times6}+\frac{33}{5\times6\times7}+\ldots\ldots\ldots$$ [4]

3. The diagram below shows the structure of an indoor farm to be built. Edges *OA*, *DC*, *IH* and *EF* are vertical, and edges *OA*, *AB*, *BC* and *CD* are equal to *EF*, *FG*, *GH* and *HI* respectively. Diagram is not drawn to scale.



Taking *O* as the origin and the vectors **i**, **j** and **k** as unit vectors along the *OD*, *OE* and *OA* respectively, the coordinates of *A* and *C* are (0, 0, 20) and (50, 0, 25) respectively. All lengths are measured in metres.

(i) Given that $-\mathbf{i} + 4\mathbf{k}$ is a vector perpendicular to the flat roof *ABGF*, and $\mathbf{i} + 2\mathbf{k}$ is perpendicular to the flat roof *BCHG*, find the obtuse angle between the two roofs. [2]

(ii) A humidity sensor is to be installed along the line *BG*, 50 metres away from *B*. Find the coordinates of the location of the sensor. [4]

(iii) A metal cable is anchored on the ground outside the farm in front of *OABCD*, at a point 25 metres away from *O* along *OD* and 10 metres in front of *OD*. The cable is attached to a point on *AB* to secure the roof *ABGF*. Find the length of the shortest cable needed. [3]

(iv) State an assumption for the above calculations to be valid. [1]

4. The function f is defined by $\mathrm{f} : x \mapsto \dfrac{1}{(x-2)^2}$ for $x < m$ where $m < 2$.

(i) Find the range of values of *m* such that both f^{-1} and f^{2} exist. [4]

(ii) Suppose the range of values of *m* found in part **(i)** holds, find the functions f^{-1} and f^{2}, and state their domains and range. [6]

Section B: Statistics [60 marks]

5. Two fair six-sided dice are thrown. The random variable X is the smaller of the two scores if they are different, and their common value if they are the same.

(i) Show that $\mathrm{P}(X=2)=\dfrac{1}{4}$ and find the probability distribution of X. [2]

(ii) Hence find $\mathrm{E}(X)$ and $\mathrm{Var}(X)$. [2]

A game is played with Ivan and Jon taking turns to throw the two dice each. Ivan throws the dice first and the player who first obtain the value of X equals to 2 wins the game. If Ivan wins the game, Jon pays him \$7. Otherwise, Ivan pays Jon \$10.

(iii) Find the player who has the higher expected gain. Justify your answer. [3]

6. **(a)** During a class reunion, 4 men and 6 women decide to stand in a row to take a class photograph. Find the number of ways that they can do so if

(i) there is no restriction, [1]

(ii) at least 2 men stand together, [2]

(iii) exactly 5 women stand together. [3]

(b) The 10 persons then sit at a round table for dinner. Find the probability that 2 particular women sit opposite each other. [2]

7. At a supermart, the option of using cashless payment for purchases is available to customers. On average, p% of the customers use cashless payment.

The manager of the supermart selects a sample of 30 customers for a survey.

(i) The probability of not more than 1 of customers surveyed use cashless payment for their purchases is 0.245. Write down an equation in terms of p. Find p, giving your answer correct to 3 decimal places. [3]

For the remaining parts of this question, use $p=30$.

(ii) Find the probability that the last customer of the 30 selected for survey is the 10^{th} customer who uses a cashless payment for his purchase. [2]

Another manager of the supermart also selects a sample of 30 customers.

(iii) Given that there are a total of less than 16 customers using cashless payment for their purchases from both samples, find the probability that one of the sample has more than 12 customers using cashless payment for their purchases. [3]

(iv) State one assumption you have made in the calculation in part **(iii)**. [1]

8. Verde wants to investigate the time taken for different volumes of water to cool to room temperature. He prepared a few samples of different volume and heated the samples to their boiling point, and then recorded the time taken for the water to cool to room temperature. The results are given in the table.

Volume (x/cm^3)	100	200	300	400	500	600	700
Time (t/min)	14	23	47	83	a	172	293

(i) It is known that the regression line of t on x is $t = -65.1429 + 0.4318x$. Show that a = 121. [2]

(ii) Draw a scatter diagram for the data and find the product moment correlation coefficient between x and t. [2]

(iii) Comment whether the regression line is appropriate based on

(a) the scatter diagram; [1]

(b) the context. [1]

Verde considers using one of the following two models:

$$\text{A: } t = a + bx^2, \quad \text{B: } t = ae^{bx},$$

where $a, b \in \mathbb{R}$, for the relationship between x and t.

(iv) Explain which is the better model and find the equation of a suitable regression line for that model. [2]

(v) Estimate the time taken for 450 cm^3 of water to cool from boiling point to room temperature. Comment on the reliability of the estimate. [3]

9. A newspaper report claims that the mean starting monthly salary of a fresh university graduate is \$3500. However a human resource manager believes that this claim is incorrect. A random sample of 60 fresh university graduates is surveyed and their starting monthly salaries, \$$x$, are summarized by

$$\sum x = 207000, \quad \sum (x - 3400)^2 = 5450000.$$

(i) Calculate the unbiased estimates of the population mean and variance. [2]

(ii) Test, at the 5% significance level, whether the human resource manager's belief is correct. [4]

(iii) State, in the context of the question, the meaning of the p-value found in part **(ii)**. [1]

A second sample of 50 fresh university graduates is surveyed and the sample mean and standard deviation of their starting salaries are found to be \$$\overline{y}$ and \$342 respectively.

(iv) Find the range of values of $\overline{y}$ such that this second sample would result in accepting the manager's belief at the 5% significance level. [3]

(v) Suppose the claim by the newspaper is correct and the standard deviation of the starting salaries of fresh university graduates is \$543, find the probability that a random sample of 50 fresh university graduates have its mean salary below \$3350. [2]

10. Gary likes to take part in duathlons where there is a total running distance of 15 km and a total cycling distance of 36 km. His timings, in minutes, for running and cycling are modelled as having independent normal distributions with means and standard deviations as shown in the table.

	Mean	Standard Deviation
Timing for running (min)	100	15
Timing for cycling (min)	μ	σ

(i) The probability that he completes his cycling in less than an hour is equal to the probability that he completes his cycling after 2 hours. Write down the value of μ. [1]

(ii) The probability that he completes his cycling under 80 minutes is 0.158655. Show that $\sigma = 10$. [2]

(iii) Find the probability that thrice the time needed for him to complete cycling is more than an hour from twice the time needed for him to complete the running. [2]

(iv) Find the probability that his timing for cycling is less than 80 minutes and his timing for running is less than 90 minutes. [2]

(v) Show that the probability of Gary finishing the duathlon under 170 minutes is 0.134. Give a reason why this probability is greater than the probability calculated in part **(iv)**. [2]

Gary's target timing for completing a duathlon is under 2 hours 50 minutes. Over the past two years, he has already completed 9 duathlons.

(vi) Find the probability that he achieves his target timing in at least 3 but fewer than 6 of the duathlons over the last two years. State an assumption needed for your calculation to be valid. [4]

END OF PAPER

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